ELEC60005
EE9-AO19

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE/EIE PART III/IV: MEng, BEng and ACGI

BIOMEDICAL ELECTRONICS

Tuesday, 10 December 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer 4 out of 5 questions

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Georgiou
Second Marker(s) : T. Constandinou

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1. This question relates to the design of a multichannel instrument for *measuring bio-potentials from muscles* located on the back. The instrument will use a 5-channel electrode system shown in Figure 1.1 and the specifications of commonly observed bio-potentials are shown in Table 1.1.

   a) The system can be implemented on a PCB with external components or an Application Specific Integrated Circuit (ASIC) in CMOS. Explain the advantages and disadvantages of each and state which you would use and why.

   [5]

   b) Design a complete instrumentation system for recoding bio-potentials from the back according to the following specifications:

   Instrumentation Amplifier Gain: 20
   Low-Pass Filter Roll off: 60dB/decade
   High-Pass Filter Roll off: 40dB/decade
   Total number of op-amps available for each filter: 1
   Supply Range: -1.8V to 1.8V

   State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 1nF, 10nF and 100nF capacitors are available for the filters.

   [10]

   c) Using your design from 1.b, sketch the schematic of the complete instrument assuming your system has only one ADC output and state any timing considerations to make sure all 5 channels are recorded in real time. You may use a single symbol to represent 1.b in your full schematic.

   [5]

Table 1.1

Signal	**Frequency range (Hz)**	**Amplitude range (mV)**
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

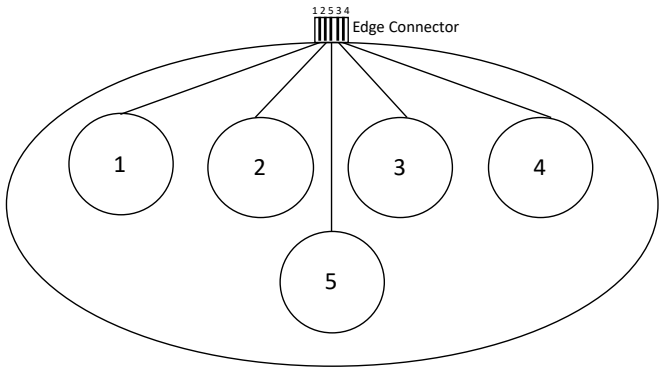


Figure 1.1

2. This question relates to electrical stimulation applied to sensory prostheses:

   a) Describe the relative merits of different electrode configurations (e.g. monopolar, bipolar, tripolar, multipolar). Why do most clinical devices utilise monopolar configurations? [5]

   b) Explain why it is challenging to apply electrical stimulation to applications requiring high resolution, for example, a retinal prosthesis for restoring vision. [5]

   c) Sketch a biphasic waveform suitable for delivering 30nC/phase stimulus for a 22-channel cochlear implant. Annotate magnitudes and timings on your sketch. State any assumptions made. [5]

   d) (i) Estimate the overall power consumption of the cochlear implant assuming an average stimulus as above. [4]

   (ii) How long would a PowerOne Implant battery last (Zinc air, 1.4V, 640mAh)? [1]

3. Figure 3.1 shows a smart contact-lens system for non-invasive measurement of tear glucose. The glucose concentration in the tear fluid is within 0.1-0.6mM for a healthy person and may go up to 1mM in patients with diabetes. The Glucose sensor comprises a working electrode (WE) with an area of $1mm^2$, a reference electrode (RE) (area:$0.3mm^2$) and a counter electrode (CE) (area:$2mm^2$). The sensor readout IC receives AC power from the antenna and sends the measured data back to the transmitter. The instrumentation operates on a single supply with a VDD of 1.8V.

The working electrode is coated with glucose oxidase to give a peak current to glucose at V_{Cell}=0.7 V according to the following reaction:

$$\text{Glucose}+O_2 \xrightarrow{Glucose\ Oxidase} \text{gluconic acid}+ H_2O_2$$
$$H_2O_2+O_2 \xrightarrow{V=0.7V} 2H^+ + 2e^-$$

a) Calculate the peak redox current for the reaction above using the Randles Equation below. You may assume the number of electrons *n* is solely given by the oxidation of H_2O_2 , the scan rate is 1V/sec and the diffusion coefficient $D=4.38x10^{-8}\ m^2/s$:

$$i_p = 2.69x10^5 \cdot n^{3/2} \cdot A \cdot D^{1/2} \cdot C \cdot v^{1/2}$$

[2]

b) Design a potentiostat and trans-impedance amplifier (TIA) with opamps and resistors to apply the cell voltage, measure the sensor current and maximize the output swing. Annotate the schematic.

[4]

c) Describe two alternative circuits to measure the sensor current, and state one advantage and one disadvantage for each with respect to the TIA in part 3.b.

[4]

d) Common mode sensor interference and current drift limit the performance of bio-sensing systems. Propose one method that can help to compensate these effects and redesign the circuit in part 3.b and one of the circuits in part 3.c to implement this method.

[4]

e) The tear glucose measurement has a non-linear dependance on pH. This however can be compensated with the addition of a sensor to measure pH. Design a complete front end for the smart-contact lens which includes a glucose and pH measurement but also compensation for common mode sensor interference and drift. You only have one ADC to sample both channels.

[6]

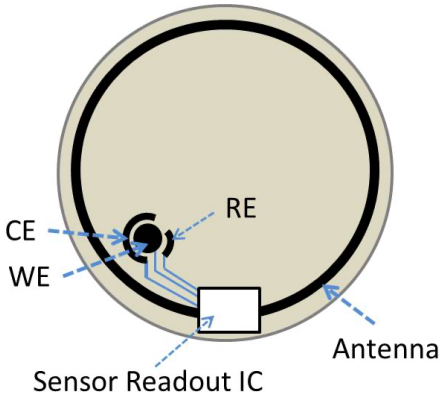


Figure 3.1

4. This question relates to implantable brain computer interface (iBCI) devices:

   a) What are the advantages and disadvantages of an intravascular BCI compared to an intracortical BCI? [5]

   b) Estimate the amount of raw data (kbps) generated by this iBCI (illustrated in Figure 4.1). [5]

   c) Explain how information is extracted from the low-level data received by the computer. [5]

   d) In what ways could such an iBCI be improved? [5]

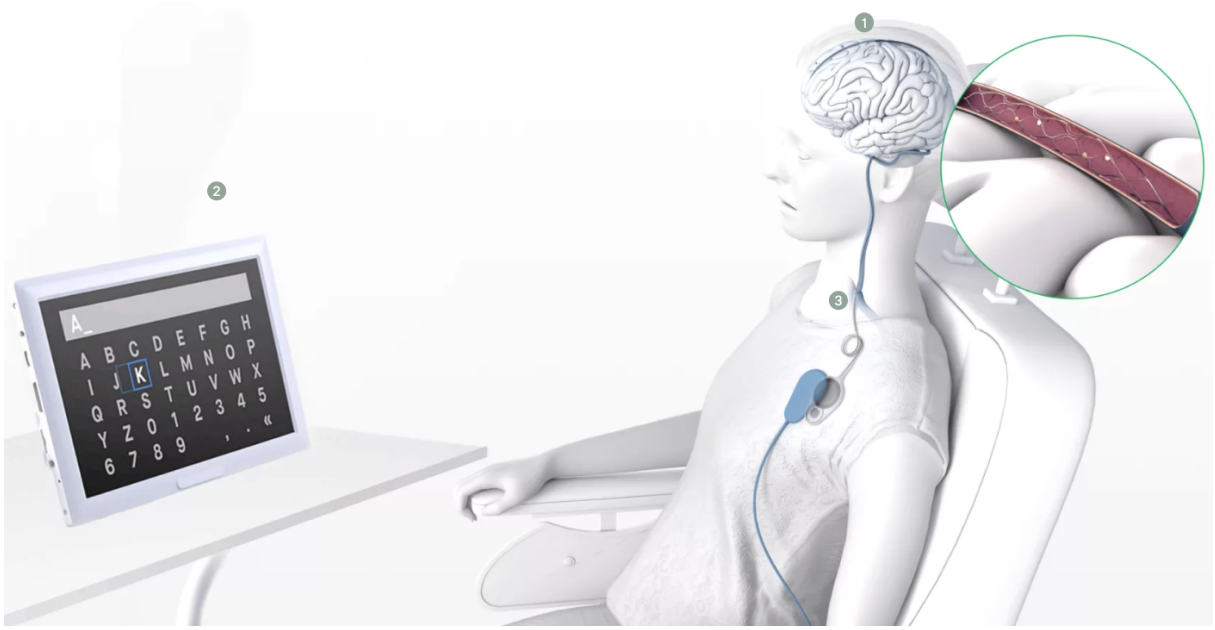


Figure 4.1. an intravascular iBCI that is currently in human trials. This utilises a 16-electrode array inserted in the superior sagittal sinus blood vessel.

5. This question relates to the design of an instrument for conducting DNA sequencing.

   a) Explain the process of DNA sequencing describing the end-to-end procedure from an unknown strand of human DNA and the sensing mechanism used. [10]

   b) DNA sequencing has been revolutionised with the introduction of semiconductor sequencing. This uses a CMOS chip with millions of chemical sensors to conduct an in-parallel investigation of the DNA. You are required to design this CMOS chip describing the following:

      - The choice of your sensor to detect DNA and the mechanism of sensing.
      - The sensors operating characteristics, showing equations describing its function.
      - The choice of instrumentation for this sensor.
      - The circuit diagram implementation for an array to scale the design to millions of sensors.

      [10]